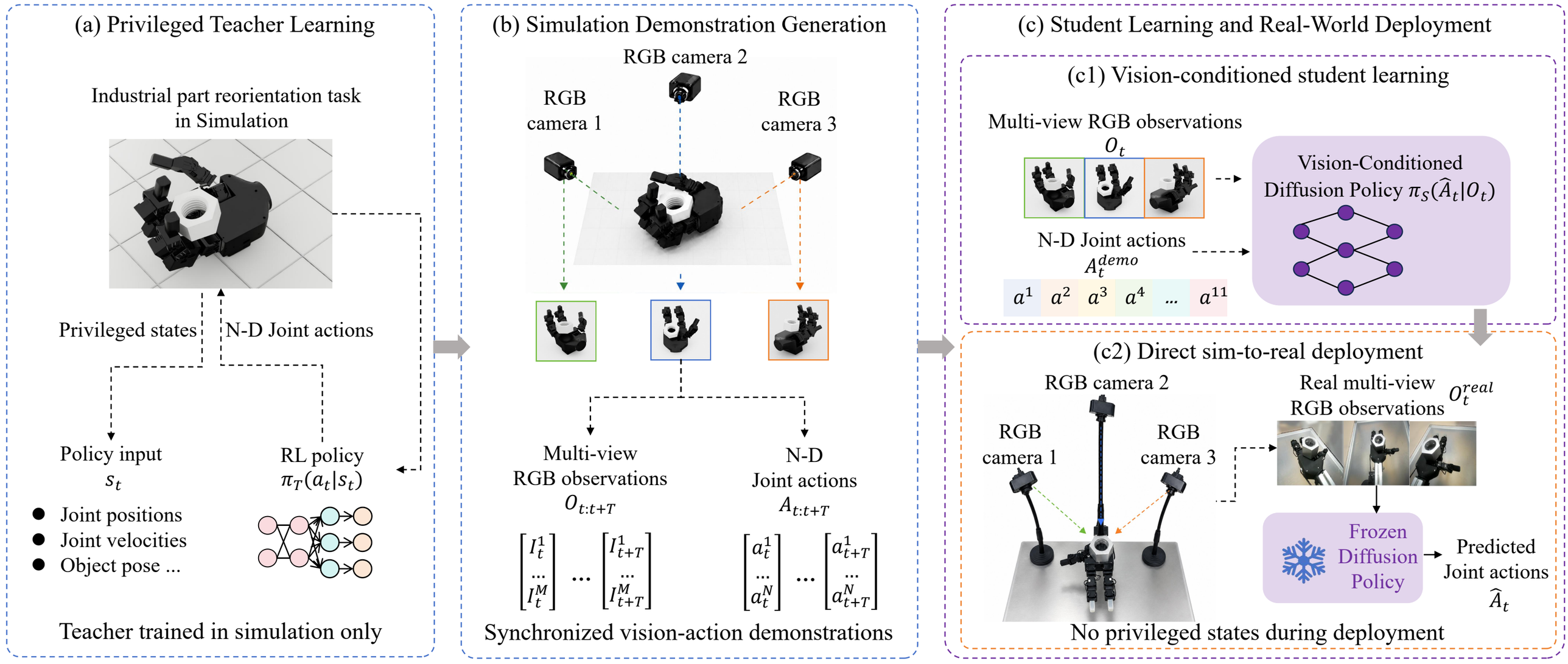

(a) Privileged Teacher Learning
Industrial part reorientation task in Simulation
Privileged states
N-D Joint actions
Policy input s_t
RL policy $\pi_T(a_t|s_t)$
Joint positions
Joint velocities
Object pose ...
Teacher trained in simulation only
(b) Simulation Demonstration Generation
RGB camera 2
RGB camera 1
RGB camera 3
Multi-view RGB observations $O_{t:t+T}$
N-D Joint actions $A_{t:t+T}$
Synchronized vision-action demonstrations
(c) Student Learning and Real-World Deployment
(c1) Vision-conditioned student learning
Multi-view RGB observations O_t
N-D Joint actions A_t^{demo}
a^1 a^2 a^3 a^4 ... a^{11}
Vision-Conditioned Diffusion Policy $\pi_S(\hat{A}_t|O_t)$
(c2) Direct sim-to-real deployment
RGB camera 2
RGB camera 1
RGB camera 3
Real multi-view RGB observations O_t^{real}
Frozen Diffusion Policy
Predicted Joint actions $\hat{A}_t$
No privileged states during deployment